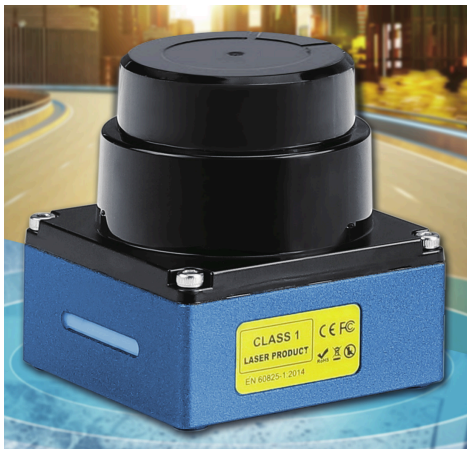


2D 360° ZONE-SWITCHING LIDAR · VF SERIES

360° ZONE OLEI VFD1-10 SWITCHING

A 300-gram full-circle scanner that carries the zone logic itself — 16 banks of three levels, four switching inputs, four outputs, NPN or PNP — and keeps detecting in light rain, direct sun and after a 15 kV air discharge.



■ WHY THIS SENSOR

- **360° scanning with zone-switching I/O** — 16 banks × 3 levels and 4 in / 4 out from a full-circle sensor, not a 270° one
- **NPN or PNP outputs, hardware selectable** — 50 mA per output — wires to the platform's existing logic either way
- **ESD hardened to Level 4 (EN IEC 61000-4-2:2009)** — 8 kV contact discharge, 15 kV air discharge — the spec that survives a dry warehouse
- **Detects in moderate or light rain** — with an anti-pollution optical window for outdoor and dock-door duty
- **80 000 lux sunlight immunity** — works facing a roller door at the wrong time of day
- **0.1–10 m, and 10 m at 10% reflectivity as well as 80%** — the 10% figure matches the 80% figure — unusual, and it matters for dark targets
- **300 g in a 65 × 65 × 70 mm body, < 5 W** — the smallest and lightest 360° sensor in the OLEI line
- **SafetyConfigTool zone setup plus OLAMView** — draw the banks in the GUI, 130 ms response at 10 Hz

0.1–10 m

WORKING DISTANCE

360°

FIELD OF VIEW

16 × 3

ZONES / LEVELS

300 g

WEIGHT

■ APPLICATION SCENARIOS



AMR & AGV

Full-circle obstacle avoidance from one chassis mount



INTRALOGISTICS

Forklifts, tuggers and conveyor zones in active aisles



AREA MONITORING

Perimeter and equipment-envelope intrusion detection



AUTONOMOUS VEHICLES

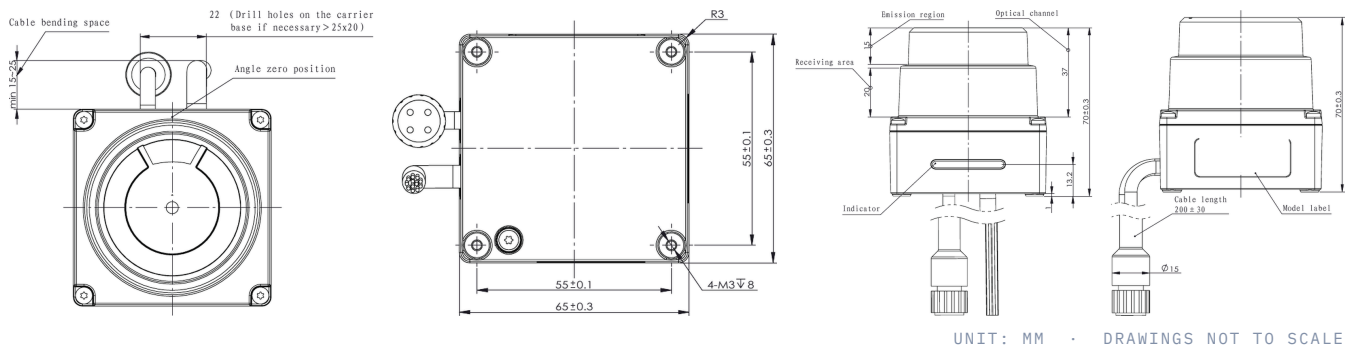
Forward perception for on- and off-road platforms

VFD1-10 — SPECIFICATIONS

PARAMETER	TECHNICAL DATA
Sensing technology	2D scanning ToF LiDAR
Laser wavelength	905 ± 20 nm
Laser class	Class 1 (IEC 60825-1:2014)
Channels	1
Field of view (horizontal)	360°
Beam divergence	8 mrad (H) 3 mrad (V)
Scanning frequency	10, 15, 25 Hz
Angle resolution	0.25° @ 10 Hz 0.5° @ 15 Hz · 1° @ 25 Hz
Working distance (based on reflectivity)	0.1–10 m @ 80% and @ 10% reflectivity 0.1–5 m @ 4% · 0.1–10 m @ reflector
Distance resolution	1 mm
Accuracy ①	Absolute < ±30 mm (0.4–10 m) Repeat ≤ 20 mm (0.4–10 m)
Point cloud density	14.4K @ 10 Hz 10.8K @ 15 Hz · 9K @ 25 Hz
Horizontal error (horizontal angle)	≤ ±1°
Response time	130 ms @ 10 Hz

PARAMETER	TECHNICAL DATA
Number of bank fields	16 banks, 3 levels/bank
Switching inputs	4
Switching outputs	4
Output mode	NPN / PNP (hardware optional)
Output current	50 mA
Power supply	12-core adapter with cable Cable length 20 cm, TPU
Communication	4-core adapter with cable, M12×1 female Cable length 200 cm, TPU
Operating voltage	12–26 V DC
Power consumption @ 25 °C	< 5 W (no load)
Mechanical dimensions	65 × 65 × 70 mm
Weight	300 g ± 5%
Degree of protection	IP65
Operating temperature	–10 to +50 °C
Storage temperature	–20 to +70 °C
Ambient light limit	80 000 lux
Function software	SafetyConfigTool
Basic software	OLAMView

OUTLINE AND MOUNTING



Certification. RoHS, CE, FCC, FDA, EMC per IEC 61000-6-2:2016-08 and IEC 61000-6-3:2006-07. ESD per EN IEC 61000-4-2:2009 Level 4 — 8 kV contact discharge, 15 kV air discharge. Vibration per IEC 60068-2-6:2007. Shock per IEC 60068-2-27:2008.

Functional safety. The manufacturer's specification for this product states laser safety Class 1 per IEC 60825-1:2014 and the EMC, vibration and shock standards listed above. It does not state an IEC 61496 Type rating or an ISO 13849 performance level. Treat the zone outputs as obstacle-avoidance and area-monitoring signals, not as a rated safety function, and do not use this sensor as the sole protective device. Where a rated scanner is required, ask MorpheusTEK about the OLEI GS1-5.

Notes. ① Standard value; actual value depends on environmental conditions, and both accuracy figures are specified from 0.4 m rather than from the 0.1 m minimum range. Angle resolution and point rate both fall as scan frequency rises — 0.25° and 14.4K pts/s are the 10 Hz case. Rain capability is stated for moderate or light rain, not driving rain or standing spray.

Installation. A 360° sensor needs its full circle clear — mount it above the platform's own structure rather than recessed into it, or the blind sectors will be created by the vehicle instead of the sensor. Allow the stated cable bending space beneath the unit; drill the carrier base if necessary. Verify every configured bank on the platform after any change to mounting height or angle, and clean the optical window on a schedule in dirty environments.

OLEI and the OLEI logo are trademarks of Hangzhou OLE-Systems Co., Ltd. MorpheusTEK is the exclusive North American distributor for OLEI LiDAR. Specifications are subject to change without notice; confirm current values with MorpheusTEK before design-in. Doc. MTK-OLEI-VFD110-2026.09 · Source: OLEI VFD1-10 2D 360° LiDAR Sensor datasheet, DOC.# DSEN-VFD1-10-202505.